

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

```
data = pd.read_csv('Iris.csv')
```

```
X = data.iloc[:, :4].values
y = data['Species'].values
print(data.head(5))
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2)
```

```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

```
from sklearn.naive_bayes import GaussianNB
classifier = GaussianNB()
classifier.fit(X_train, y_train)
```

```
GaussianNB()
```

```
y_pred = classifier.predict(X_test)
```

```
from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test, y_pred)
from sklearn.metrics import accuracy_score
print("Accuracy : ", accuracy_score(y_test, y_pred))
print(cm)
```

```
Accuracy : 0.9666666666666667
[[10  0  0]
 [ 0 10  0]
 [ 0  1  9]]
```

```
df = pd.DataFrame({'Real Values':y_test, 'Predicted Values':y_pred})
```

```
print(df)
```

	Real Values	Predicted Values
0	Iris-versicolor	Iris-versicolor
1	Iris-versicolor	Iris-versicolor
2	Iris-virginica	Iris-virginica
3	Iris-virginica	Iris-virginica
4	Iris-setosa	Iris-setosa
5	Iris-setosa	Iris-setosa
6	Iris-virginica	Iris-virginica
7	Iris-virginica	Iris-virginica
8	Iris-versicolor	Iris-versicolor
9	Iris-virginica	Iris-virginica
10	Iris-versicolor	Iris-versicolor
11	Iris-virginica	Iris-versicolor
12	Iris-setosa	Iris-setosa
13	Iris-versicolor	Iris-versicolor
14	Iris-setosa	Iris-setosa
15	Iris-virginica	Iris-virginica
16	Iris-versicolor	Iris-versicolor
17	Iris-versicolor	Iris-versicolor
18	Iris-virginica	Iris-virginica
19	Iris-versicolor	Iris-versicolor
20	Iris-setosa	Iris-setosa
21	Iris-virginica	Iris-virginica
22	Iris-versicolor	Iris-versicolor
23	Iris-setosa	Iris-setosa
24	Iris-virginica	Iris-virginica
25	Iris-setosa	Iris-setosa
26	Iris-setosa	Iris-setosa
27	Iris-setosa	Iris-setosa
28	Iris-setosa	Iris-setosa
29	Iris-versicolor	Iris-versicolor

```
from sklearn.metrics import precision_score,  
recall_score,accuracy_score
```

```
m=accuracy_score(y_test,y_pred)
```

```
print("error rate:-",1-m)
```

```
error rate:- 0.033333333333333326
```

```
print('Precision:',precision_score(y_test,y_pred,average='micro'))
```

```
Precision: 0.9666666666666667
```

```
print("Recall Score:",recall_score(y_test,y_pred,average='micro'))
```

```
Recall Score: 0.9666666666666667
```